

Google Colab: Get the Cloud on Your Desktop



Here's how you can use Google Colab to bring cloud computing to your PC.

IaaS (Infrastructure as a Service), PaaS (Platform as a Service) and SaaS (Software as a Service) have been the three major cloud computing services for a while now. However, the growing needs of computing, including faster connectivity, have given rise to a new cloud computing model called DaaS or Desktop as a Service. Though this model has been around as virtual desktop infrastructure (VDI) for some time, it was formalised in 2021 with the introduction of Windows 365 cloud PCs by Microsoft.

Fast forward to today. Cloud PCs are still relatively expensive for the average user. Also, Windows 365 is built for business and enterprise customers only. This is where Google Colab comes in. It offers temporary access to containers using IPython notebooks. The free tier offers around 12 hours of runtime, which is usually enough for

most purposes. It also offers around 1 Gbps internet speed and can be used for temporary tasks. For persistent file storage, a Google Drive can be mounted, with the free tier providing around 15GB of storage.

Hence, it makes good sense to use Colab containers as virtual machines for cloud PC. This can be achieved by installing the essential applications and a desktop environment on the container.

Let us start by creating an IPython Notebook and installing the required applications, which are: a web browser, a desktop environment, and a remote desktop server. Since the goal is to run the notebook inside Colab, which is on the browser, a proxy is needed to render the remote desktop on the browser itself.

Start by installing Firefox as the browser, XFCE as the desktop environment, VNC server for remote desktop,